

Product Requirements Document (PRD)

Sleekandchic by Salma – E-commerce Platform

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1. Product Overview

Sleekandchic by Salma is a women’s fashion brand that sells clothing items with size and color variations. The platform must enable a frictionless shopping experience for guests and registered users, support multi-channel order capture (website, POS, social media), and provide a custom shipping-price engine based on state/region configuration. The unique selling proposition is exceptional customer service, reinforced by real-time order notifications via email and optional WhatsApp.

The system will be a web-based application with a mobile-first responsive UI, a secure admin dashboard, and RESTful APIs for frontend-backend communication and third-party integrations (payment gateways, WhatsApp business API, email service).

2. Functional Requirements

ID	Requirement	Description	Priority
FR1	Product Catalog	Users can browse products, filter by category, view product detail pages showing images, description, price, and available variants (size, color).	High
FR2	Variant Selection	On product detail page, users select size and color (if applicable) before adding to cart. Stock availability is validated in real-time.	High
FR3	Shopping Cart	Persistent cart (guest cart stored in localStorage or server-side session). Users can add, update quantity, remove items, and view subtotal.	High
FR4	Guest Checkout	Users can checkout without creating an account. Required fields: email, phone, shipping address (including state/region), and payment method.	High
FR5	Payment Methods	Integration with card payment gateway (e.g., Stripe/PayPal) and manual bank transfer option. After successful payment, order status moves to “Processing”.	High
FR6	Order Creation	Order is created immediately after payment confirmation. Order contains: customer info, items (with variant SKUs), quantities, prices, taxes, shipping cost, total, and timestamps.	High

ID	Requirement	Description	Priority
FR7	Order Confirmation	Customer receives an email confirmation (and optional WhatsApp message) with order number and summary. Business owner and assigned staff receive notification.	High
FR8	Shipping Configuration (Admin)	Admin can add/update states/regions and assign a flat shipping price per state. Shipping methods: Authorized Car Park (cheaper) or Delivery Company.	High
FR9	Dynamic Shipping Price at Checkout	During checkout, if the entered state/region exists in the shipping table, the corresponding price is shown and added to order total. If not, user selects "Other", enters state manually, and shipping fee is \$0 (to be followed up by admin).	High
FR10	Multi-Channel Order Capture	The system must accept orders from: website (FR4), POS (manual order creation by staff), and social media (manual order creation). Inventory is decremented accordingly.	High
FR11	Manual Order Creation (Admin/Staff)	From the admin dashboard, staff can create an order on behalf of a customer, select items, apply discounts, enter shipping address, and choose payment method (card or bank transfer). Stock is updated in real-time.	High
FR12	Inventory Management	Inventory is tracked at the SKU level (product + size + color). Automatic decrement on order creation (website, POS, manual). Admin can manually adjust stock (e.g., returns, damages). Low-stock alerts (configurable threshold) are shown in dashboard.	High
FR13	Discounts & Promotions	Admin can create coupon codes (fixed amount or percentage) with usage limits, start/end dates, and applicability (entire order, specific products). Coupon validation occurs at cart/checkout.	Medium
FR14	Content Management	Admin can create/publish blog posts and pages (e.g., Shipping Policy) via a WYSIWYG editor. Content is accessible via public routes.	Low
FR15	Promotional Carousel	Admin can upload images, set target URL, and define display order for a carousel shown on the homepage. Carousel items can be enabled/disabled.	Medium
FR16	Notifications System	Mandatory email notifications for: new order (customer, owner, staff), order status changes (optional). Optional WhatsApp notifications via WhatsApp Business API for order confirmation to customer and alert to admin.	High
FR17	Role-Based Access Control (RBAC)	Two roles: Admin (full system access) and Staff (limited to orders, customers, product view, manual order creation). Staff cannot modify system settings, shipping configuration, or user roles.	High

ID	Requirement	Description	Priority
FR18	Admin Dashboard	Provides: sales overview (revenue, trends, top products), order list with filter/search, order detail view, product/variant management, inventory levels, shipping configuration UI, coupon management, customer list, carousel management, content/blog management.	High
FR19	Responsive Design	UI must be mobile-first, responsive, and follow the visual style of https://knanfe.com/ (clean, minimal, professional).	Medium
FR20	Performance	Page load time < 2 seconds on 3G for homepage and product listing; API response time < 200ms for 95th percentile.	High
FR21	Security	All admin endpoints require authentication and authorization. Passwords hashed with bcrypt. HTTPS enforced. Input validation and output encoding to prevent XSS/SQLi. CSRF tokens on state-changing requests.	High
FR22	Analytics & Reporting	Basic sales reports exportable as CSV (daily, weekly, monthly). Google Analytics integration for storefront.	Low

3. User Stories

Customer (Guest/User)

- **US1:** As a guest, I want to browse products so I can see what's available.
- **US2:** As a guest, I want to select size and color so I can order the correct variant.
- **US3:** As a guest, I want to add items to my cart and view the cart so I can review my selection.
- **US4:** As a guest, I want to checkout without creating an account so I can complete purchase quickly.
- **US5:** As a guest, I want to see shipping cost based on my state so I know the total cost before paying.
- **US6:** As a guest, if my state is not in the list, I want to select "Other" and type my state so I can still place an order (shipping to be arranged later).
- **US7:** As a customer, I want to pay with card or bank transfer so I can choose my preferred payment method.
- **US8:** As a customer, I want to receive an email (and optionally WhatsApp) confirmation after placing an order so I have proof of purchase.
- **US9:** As a customer, I want to view a simple, professional homepage with a carousel so I can see promotions and featured products.

Staff / Admin

- **US10:** As staff, I want to log into the admin dashboard so I can perform my duties.
- **US11:** As staff, I want to view and manage orders (search, filter, see details) so I can fulfill them.
- **US12:** As staff, I want to create an order manually (POS or social media) so I can capture offline sales.
- **US13:** As staff, I want to see real-time inventory levels so I know what's in stock.
- **US14:** As admin, I want to configure shipping prices per state so I can control costs.
- **US15:** As admin, I want to add, edit, enable/disable carousel banners so I can run promotions.
- **US16:** As admin, I want to create coupon codes so I can run discount campaigns.
- **US17:** As admin, I want to manage blog content and pages so I can communicate with customers.

- **US18:** As admin, I want to view sales reports and trends so I can make business decisions.
- **US19:** As admin, I want to manage staff accounts (create, reset password, assign role) so I can control access.

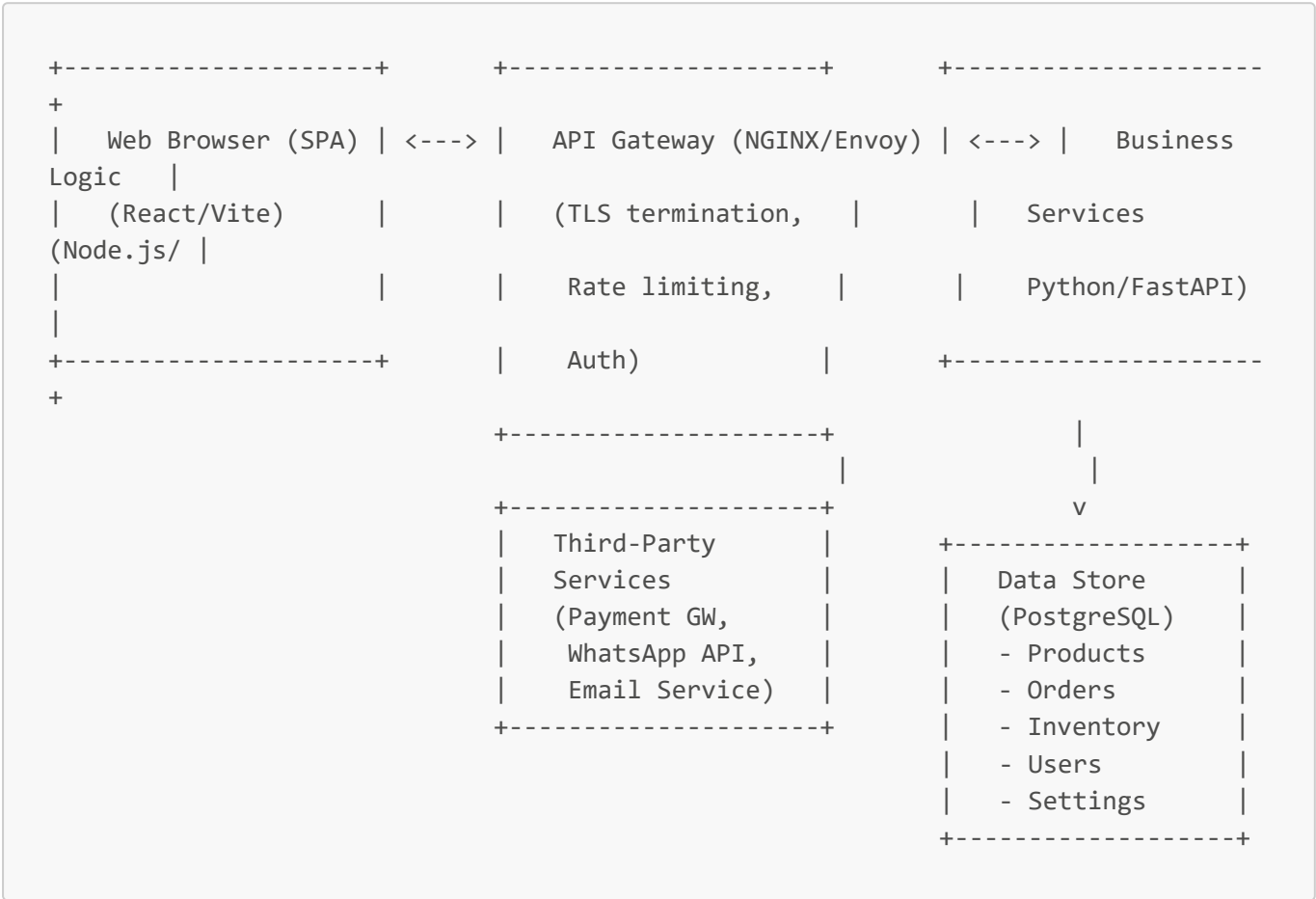
4. Non-Functional Requirements

Category	Requirement	Details
Performance	Page Load	Homepage & category pages < 2s on 3G; product detail < 1.5s; API < 200ms 95th percentile.
Scalability	Horizontal	Stateless backend services; database read replicas; CDN for static assets.
Availability	Uptime	99.9% monthly uptime SLA (excluding scheduled maintenance).
Security	Authentication	JWT-based auth for admin panel; refresh token rotation; HTTP-only cookies.
	Authorization	RBAC enforced at middleware level.
	Data Protection	TLS 1.2+ everywhere; PCI-DSS compliance for card data (use tokenization via gateway).
	Input Validation	Server-side validation for all inputs; sanitization to prevent XSS/SQLi.
	Rate Limiting	API endpoints protected against brute-force (login, coupon validation).
Reliability	Error Handling	Graceful degradation; informative error messages; logging to centralized system (ELK or similar).
	Backup	Daily automated DB backups; point-in-time recovery.
	Monitoring	Health checks, uptime alerts, APM (e.g., New Relic).
Usability	Accessibility	WCAG 2.1 AA compliance (keyboard navigation, ARIA labels, contrast).
	Language	English (primary); UI strings externalized for future i18n.
	Mobile-First	Breakpoints: ≤480px (mobile), 481-768px (tablet), >768px (desktop).
Maintainability	Code Style	ESLint + Prettier (JS/JSX); Pylint (Python) if applicable; consistent naming; modular architecture.
	Documentation	API docs (OpenAPI/Swagger); developer README; deployment guides.
Deployability	CI/CD	Automated tests (unit, integration) on pull request; staging environment; blue-green or rolling deployments.

Category	Requirement	Details
	Environment Variables	Config via .env (dev) or secret manager (prod).

5. System Architecture

High-Level Components



Description

- Frontend** – Single Page Application built with React (or Vue) and Vite, consuming RESTful APIs. Mobile-first CSS using Tailwind or similar.
- API Gateway** – NGINX (or Envoy) terminates TLS, performs basic rate limiting, routes `/api/*` to backend services, serves static assets.
- Backend Services** – Stateless microservices (or modular monolith) handling:
 - Product Service** – catalog, variants, inventory.
 - Order Service** – cart, checkout, payment integration, order lifecycle.
 - User/Auth Service** – admin/staff authentication, RBAC, session management.
 - Shipping Service** – state/region pricing logic.
 - Notification Service** – email and WhatsApp dispatch.
 - Content/CMS Service** – blog, pages, carousel management.
 - Analytics/Reporting Service** – sales aggregation. Services communicate via internal REST/gRPC or message queue (RabbitMQ/Kafka) for async tasks (e.g., sending notifications).

4. **Database** – PostgreSQL as primary relational DB. Tables designed for ACID compliance; read replicas for reporting queries.
5. **Caching** – Redis for session store (guest cart), API response caching (product listings), and rate-limit counters.
6. **File Storage** – Amazon S3 (or compatible) for product images, carousel banners, blog assets; served via CDN.
7. **Third-Party Integrations** –
 - Payment Gateway: Stripe API (card) + custom webhook for bank transfer confirmation.
 - WhatsApp: WhatsApp Cloud API (template messages for order confirmation).
 - Email: SendGrid / SES (transactional emails).
 - Analytics: Google Analytics 4 (frontend) + internal sales dashboards.

Data Flow (Checkout)

1. Customer adds items to cart (stored in Redis with TTL).
2. On checkout, frontend sends `/api/orders` with cart items, shipping address, payment method.
3. Order Service validates stock, calculates taxes, fetches shipping price from Shipping Service (based on state).
4. If payment method = card → create PaymentIntent via Stripe, return `client_secret` to frontend; after Stripe confirms, webhook updates order to `paid`. If bank transfer → order status = `pending_payment`, admin notified to confirm receipt.
5. Once paid, Order Service:
 - Decrements inventory.
 - Creates order record.
 - Triggers Notification Service (email + optional WhatsApp).
 - Returns order confirmation to frontend.
6. Frontend displays thank-you page with order number.

6. Database Design

Entity-Relationship Overview (textual)

- **Product** (id, name, description, base_price, sku, active, created_at, updated_at)
- **Variant** (id, product_id, size, color, sku_suffix, stock_quantity, active)
 - Composite unique: (product_id, size, color)
- **Order** (id, order_number, customer_email, customer_phone, status, subtotal, tax, shipping_cost, discount_amount, total_amount, payment_method, payment_transaction_id, created_at, updated_at)
- **OrderItem** (id, order_id, product_id, variant_id, quantity, unit_price, total_price)
- **StateShipping** (id, state_name, shipping_method, price, active) – stores Authorized Car Park and Delivery Company options per state.
- **Coupon** (id, code, type, value, min_order_amount, usage_limit, used_count, starts_at, ends_at, active)
- **User** (id, email, password_hash, role (admin/staff), active, created_at, updated_at)
- **StaffSession** (id, user_id, token, expires_at) – for admin dashboard auth.
- **Cart** (id, session_id (guest) or user_id, product_id, variant_id, quantity, updated_at) – stored in Redis for performance.
- **NotificationLog** (id, recipient, channel, template, status, sent_at, error_message)
- **BlogPost** (id, title, slug, content_html, published_at, active)

- **CarouselItem** (id, image_url, target_url, alt_text, display_order, active)

Key Points

- Inventory is stored at variant level; stock decrement is performed inside a transaction with order creation.
- Shipping cost is looked up via **StateShipping** where **state_name** matches the customer's state (case-insensitive trim). If not found, shipping cost = 0 and a flag **needs_shipping_review** is set on the order.
- Discounts applied at order level; coupon validation checks usage limits and date windows.
- All tables have **created_at** and **updated_at** timestamps; soft deletes via **active** flag where appropriate.
- Indexes:
 - **Product(active), Variant(product_id, active), OrderItem(order_id), Order(status, created_at), StateShipping(state_name, active), Coupon(code, active), User(email).**

7. API Design

All APIs are RESTful, JSON-encoded, served under **/api/v1**. Authenticated endpoints require **Authorization: Bearer <jwt>** cookie/header.

Public (Customer-Facing)

Method	Endpoint	Description	Auth
GET	/api/v1/products	List products with optional filters (category, search, pagination)	None
GET	/api/v1/products/{id}	Get product detail including variants	None
GET	/api/v1/variants/{id}	Get variant detail (stock, price)	None
POST	/api/v1/cart/items	Add item to cart (product_id, variant_id, quantity)	None (guest cart via session cookie)
GET	/api/v1/cart	Retrieve current cart	None
PUT	/api/v1/cart/items/{itemId}	Update quantity	None
DELETE	/api/v1/cart/items/{itemId}	Remove item	None
POST	/api/v1/orders	Create order from cart (shipping address, payment method)	None
GET	/api/v1/orders/{orderNumber}	Get order details (customer-only)	None (order number as token)

Method	Endpoint	Description	Auth
POST	/api/v1/payments/stripe/webhook	Stripe webhook (payment succeeded/failed)	None (signature verification)
GET	/api/v1/shipping/states	List all configured states (for dropdown)	None
GET	/api/v1/content/pages/{slug}	Get static page (e.g., shipping-policy)	None
GET	/api/v1/blog/posts	List published blog posts	None
GET	/api/v1/blog/posts/{slug}	Get single blog post	None
GET	/api/v1/carousel	Get active carousel items (ordered)	None

Admin/Staff (Authenticated)

Method	Endpoint	Description	Roles
POST	/api/v1/auth/login	Email/password login → returns JWT	staff/admin
POST	/api/v1/auth/logout	Invalidate token	staff/admin
GET	/api/v1/dashboard/summary	Sales overview (revenue, orders today, top products)	admin
GET	/api/v1/orders	List orders with filters (status, date range, search)	staff/admin
GET	/api/v1/orders/{id}	Get order detail	staff/admin
PUT	/api/v1/orders/{id}/status	Update order status (processing, shipped, delivered, cancelled)	staff/admin
POST	/api/v1/orders/manual	Create manual order (staff)	staff/admin
GET	/api/v1/products (admin)	List all products (incl. inactive)	admin
POST	/api/v1/products	Create product	admin
PUT	/api/v1/products/{id}	Update product	admin
DELETE	/api/v1/products/{id}	Soft-delete product	admin

Similar CRUD endpoints for **variants**, **states****shipping**,

Method	Endpoint	Description	Roles
coupons, users (staff only view), blogposts, carouselitems.			
GET	/api/v1/notifications/logs	View notification dispatch logs (admin)	admin
POST	/api/v1/notifications/test	Send test email/WhatsApp (admin)	admin

Error Format

```
{
  "error": {
    "code": "VALIDATION_ERROR",
    "message": "Quantity must be greater than zero",
    "details": { "field": "quantity", "received": 0 }
  }
}
```

Versioning

- Use URL versioning ([/api/v1/](#)). Increment major version for breaking changes.

8. UI/UX Guidelines

Visual Style

- **Inspiration:** <https://knanfe.com/> – clean, minimal, generous white space, professional typography.
- **Color Palette:** Primary – muted rose or burgundy (#C08081); Secondary – soft grey (#F5F5F5); Accent – dark charcoal (#3333); Background – white.
- **Typography:** Headings – Playfair Display (serif); Body – Inter or Helvetica Neue (sans-serif). Web-safe fallback.
- **Iconography:** Simple line icons (e.g., Feather Icons) for actions (cart, search, user).

Layout

- **Header:** Logo left, navigation (Home, Shop, Blog, About, Contact) right, cart icon with badge.
- **Hero Section:** Full-width banner with tagline, CTA button (“Shop New Arrival”).
- **Product Listing:** Grid (2-col mobile, 3-col tablet, 4-col desktop). Each card shows image, name, price, variant selector (size/color dropdowns), “Add to Cart” button.
- **Product Detail:** Image carousel (thumbnails), name, price, description, variant selection (size/color swatches), stock indicator, “Add to Cart”.
- **Cart Page:** List items with thumbnail, name, variant, qty, price, line total; summary (subtotal, shipping, tax, discount, total); checkout button.

- **Checkout Flow** (single page or wizard):
 1. Shipping Information (email, phone, address – state autocomplete with “Other” option).
 2. Shipping Method & Cost (displayed automatically based on state; if “Other” → \$0 and note about follow-up).
 3. Payment Method (Card via Stripe Elements, Bank Transfer instructions).
 4. Review & Place Order.
- **Order Confirmation Page:** Order number, summary, “Continue Shopping” button, note that email/WhatsApp sent.
- **Admin Dashboard:**
 - Sidebar navigation (Dashboard, Orders, Products, Inventory, Shipping, Coupons, Customers, Content, Carousel, Users).
 - Main area: cards for KPIs (total sales, orders today, average order value, low stock count).
 - Tables with search, pagination, bulk actions (e.g., change order status).
 - Forms use modal dialogs or inline editing with validation.

Interactions

- **Autocomplete for State:** As user types, suggest from configured states; if not found, allow manual entry and show “Other” selection.
- **Variant Selection:** Swatches for color; dropdown for size; disabled swatch if out of stock (tooltip: “Out of stock”).
- **Cart Persistence:** Guest cart saved in browser `localStorage`; on login, merge with server cart (if applicable).
- **Notifications:** Toast messages for add-to-cart success/removal; inline validation errors.
- **Loading States:** Skeleton loaders for product lists; spinner for API calls.
- **Accessibility:** All interactive elements keyboard navigable; ARIA labels for icons; sufficient contrast (AA); focus outlines visible.

Responsive Breakpoints

Width	Layout
< 480px	Single column, hamburger menu
480-768px	Two-column product grid, sidebar collapsed
> 768px	Full sidebar, multi-column grids

Performance Optimizations

- Lazy-load images below viewport.
- Use `srcset` for responsive images.
- Minify CSS/JS; leverage HTTP/2.
- Serve static assets via CDN with cache-control.
- Enable browser caching for assets (1 year hash-based filenames).

9. Assumptions & Constraints

ID	Assumption / Constraint	Rationale
A1	Payment gateway supports webhook for asynchronous confirmation (Stripe).	Needed to update order status after payment without blocking checkout.
A2	WhatsApp Business API account is available and approved for template messages.	Required for optional WhatsApp notifications.
A3	Inventory is managed at SKU level (product+size+color).	Enables accurate stock tracking for variants.
A4	Tax calculation is not required (or flat rate) – if needed, can be added later as a tax service.	Simplifies initial scope; tax rules can be jurisdiction-specific.
A5	Shipping price is flat per state (no weight- or distance-based).	Matches business description of assigning price per state.
A6	Admin can manually enter shipping cost for “Other” states after order is placed.	Aligns with requirement: order proceeds without shipping fee; admin follows up.
A7	Storefront will be hosted on a domain with SSL; internal services communicate over TLS or within a VPC.	Security requirement.
A8	Expected order volume: up to 500 orders/day initially, scaling to 5k/day.	Inform sizing of DB and caching layer.
A9	No third-party marketplace integration (e.g., Amazon, eBay) at launch.	Keeps scope focused.
A10	Content (blog, pages) is authored by admin; no user-generated comments.	Simpler CMS.
C1	Must support PCI-DSS SAQ-D via tokenization; we will not store raw card data.	Mandatory for card payments.
C2	System must be operable with a small admin team (≤ 5 users).	Influences UI simplicity.
C3	Deployment environment: Linux servers (Ubuntu 22.04) with Docker Compose for dev, Kubernetes (or managed service) for prod.	Constrains infrastructure choices.

10. Future Enhancements

ID	Enhancement	Description
FE1	Loyalty Program	Points per purchase, redeemable for discounts.
FE2	Advanced Shipping	Real-time rates from carriers (DHL, FedEx) based on weight/dimensions; support for international customs fees.

ID	Enhancement	Description
FE3	Multi-Currency & Language	Support for USD, EUR, GBP; Arabic/English toggle.
FE4	Marketplace Integrations	Connect to Instagram Shopping, Facebook Shops, Amazon, etc.
FE5	AI-Powered Recommendations	"Customers also bought" based on purchase history.
FE6	Subscription Model	Recurring deliveries for certain apparel (e.g., socks, basics).
FE7	Advanced Analytics	Cohort analysis, funnel visualization, A/B testing for promotions.
FE8	POS Hardware Integration	Barcode scanner, receipt printer, cash drawer via USB/Bluetooth.
FE9	Gift Cards & Store Credit	Ability to sell and redeem gift cards.
FE10	Returns & Refunds Portal	Self-service return initiation, label generation, restocking.
FE11	Inventory Forecasting	Predictive reorder alerts based on sales velocity.
FE12	Progressive Web App (PWA)	Offline browsing, push notifications, home-screen install.
FE13	Social Login	Apple, Google, Facebook login for faster checkout (optional).
FE14	Subscription Email Newsletter	Mailchimp integration for promotional campaigns.
FE15	Environmental Impact Metrics	Show carbon footprint per order, offer eco-friendly shipping options.

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